

Supplementary Information

A unified control equation for cell size homeostasis across domains of life

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S1 First-principles derivation of the Hill-type hazard function

S1.1 Molecular setup and concentration-dilution sensor

Consider a cell of volume $V(t)$ growing exponentially at rate μ . The cell contains a division-inhibiting molecule I (e.g., Whi5 in *S. cerevisiae*, Rb in mammalian cells) whose total amount is synthesized once per cell cycle: $I_{\text{tot}} = \text{const}$. As the cell grows, the inhibitor concentration decreases:

$$[I](V) = \frac{I_{\text{tot}}}{V} \quad (\text{S1})$$

Division is promoted by a cyclin-CDK activator complex A at constant concentration $[A]_0$. The activator targets respond cooperatively to the ratio of activator to inhibitor signals through n cooperative binding sites:

$$f_{\text{active}}(V) = \frac{[A]_0^n}{[A]_0^n + [I]^n} = \frac{V^n}{V^n + K^n} \quad (\text{S2})$$

where $K \equiv I_{\text{tot}}/[A]_0$ is the volume at which inhibitor and activator signals balance.

The instantaneous division rate (hazard) is f_{active} times a maximal division attempt rate μr :

$$h(V) = \mu r \frac{V^n}{V^n + K^n} \quad (\text{S3})$$

Physical interpretation of parameters: K = volume at half-maximal activation, set by inhibitor dosage ($K = I_{\text{tot}}/[A]_0$), scales with ploidy and gene dosage; n = cooperativity of the molecular switch (number of phosphorylation sites or binding events); r = maximal division rate in units of μ^{-1} .

S1.2 Closed-form survival function

The survival function admits a closed form via the substitution $u = V^n + K^n$:

$$S(V | V_b) = \exp\left(-r \int_{V_b}^V \frac{V'^{n-1}}{V'^n + K^n} dV'\right) = \left(\frac{V_b^n + K^n}{V^n + K^n}\right)^{r/n} \quad (\text{S4})$$

The division volume density is:

$$p(V_d | V_b) = \frac{r V_d^{n-1}}{V_d^n + K^n} \left(\frac{V_b^n + K^n}{V_d^n + K^n}\right)^{r/n} \quad (\text{S5})$$

S1.3 Justification of the μ prefactor

The factor μ ensures dimensional consistency and captures the empirical growth-rate dependence of division. In *E. coli*, chromosome replication initiation is triggered by DnaA accumulation, whose production rate is proportional to growth rate[?]. The division machinery itself (FtsZ ring in bacteria, cytokinetic ring in eukaryotes) is growth-rate dependent.

S2 Existence and uniqueness of the stationary size distribution

Proposition 1 (Existence of stationary distribution). *For any $n > 0$, $K > 0$, $r > 0$, the iterated map $V_b^{(g+1)} = V_d^{(g)}/2$, where $V_d^{(g)}$ is drawn from $p(V_d | V_b^{(g)})$, admits a stationary distribution $\pi(V_b)$ on $(0, \infty)$.*

Proof sketch. The transition kernel $T(V'_b | V_b) = 2p(2V'_b | V_b)$ maps the space of probability distributions on $(0, \infty)$ to itself. We verify:

(i) **Tightness.** For any V_b , the division distribution $p(V_d | V_b)$ has support on $[V_b, \infty)$ with exponentially decaying tail (since $S(V_d) \leq (V_b/V_d)^{r/2}$ for all parameter values). Thus the sequence of iterates $\{V_b^{(g)}\}$ is tight.

(ii) **Irreducibility.** For any two intervals (a, b) and (c, d) with $0 < a < b$ and $0 < c < d$, there exists g^* such that $P(V_b^{(g^*)} \in (c, d) | V_b^{(0)} \in (a, b)) > 0$. This follows because the division density $p(V_d | V_b) > 0$ for all $V_d > V_b$.

By the Krylov–Bogolyubov theorem, tightness implies the existence of at least one invariant measure. □

Proposition 2 (Uniqueness and geometric ergodicity). *The stationary distribution π is unique, and the iterates converge geometrically: there exist constants $C > 0$ and $\rho \in (0, 1)$ such that $d_W(\pi^{(g)}, \pi) \leq C\rho^g$ for any initial distribution $\pi^{(0)}$, where d_W is the Wasserstein-1 distance.*

Proof sketch. We establish a contraction in the log-volume coordinate $u = \ln V$. The transition kernel in u -space has the form $u' = \ln(V_d/2)$ where V_d is drawn from a distribution that concentrates near $V_b \cdot 2^{1/r}$ (the median). For the adder regime ($n = 1$, $K \gg V_b$), the map $\langle u' \rangle \approx \ln(K \ln 2 / (2r))$ is independent of u , giving strong contraction. For the sizer regime ($n \rightarrow \infty$), $\langle u' \rangle \approx \ln(K/2)$, again independent of u . In general, $\partial \langle u' \rangle / \partial u < 1$ for all parameter values, establishing the contraction condition for Harris’s theorem. □

S3 Detailed limiting-case derivations

S3.1 Sizer limit ($n \rightarrow \infty$)

For $V < K$: $V^n/K^n = (V/K)^n \rightarrow 0$, so $h(V) \rightarrow 0$.

For $V > K$: $V^n/(V^n + K^n) = 1/(1 + (K/V)^n) \rightarrow 1$, so $h(V) \rightarrow \mu r$.

The survival function for $V \geq K$:

$$S(V) = \exp\left(-r \int_K^V \frac{dV'}{V'}\right) = \left(\frac{K}{V}\right)^r \quad (\text{S6})$$

As $r \rightarrow \infty$, $S(V) \rightarrow 0$ for any $V > K$, and the division distribution becomes $p(V_d) \rightarrow \delta(V_d - K)$. Division size is K regardless of V_b : **pure sizer**.

Error estimate: For large but finite n , the transition width from $h \approx 0$ to $h \approx \mu r$ scales as $\delta V/K \sim 1/n$ (from the derivative of the Hill function at $V = K$). The CV of division size due to this finite sharpness is $\text{CV} \sim 1/(nr)$.

S3.2 Timer limit ($n \rightarrow 0^+$)

As $n \rightarrow 0^+$, for any $V > 0$: $V^n = e^{n \ln V} \rightarrow 1$ and $K^n \rightarrow 1$. Therefore:

$$h(V) \rightarrow \mu r \cdot \frac{1}{1+1} = \frac{\mu r}{2} \quad (\text{S7})$$

This is a constant hazard (independent of V). Converting to time using $V = V_b e^{\mu t}$:

$$S(t) = \exp\left(-\frac{r}{2} \mu t\right) \quad (\text{S8})$$

This is an exponential distribution with rate $\lambda = \mu r/2$. The mean interdivision time $\langle T \rangle = 2/(\mu r)$ is **independent of birth size**: a stochastic timer.

Important caveat: This is a *memoryless* (exponential) timer, not a deterministic timer with fixed T . A deterministic timer would require a delta-function distribution in T , which cannot arise from a constant hazard. The exponential timer has $\text{CV}(T) = 1$, whereas a deterministic timer has $\text{CV}(T) = 0$. Biological timers (e.g., light-cycle control in *Chlamydomonas*?) may involve additional clock mechanisms not captured by the hazard framework.

Error estimate: For small but finite n , the hazard acquires a weak size dependence: $h(V) \approx (\mu r/2)(1 + n \ln(V/K)/2 + O(n^2))$. The induced correlation between T and V_b scales as $|\text{Corr}(T, V_b)| \sim n/2$.

S3.3 Adder limit ($n = 1, K \gg \langle V_b \rangle$)

With $n = 1$, the survival function becomes:

$$S(V | V_b) = \left(\frac{V_b + K}{V + K} \right)^r \quad (\text{S9})$$

The division density is a shifted Pareto-like distribution:

$$p(V_d | V_b) = \frac{r}{V_d + K} \left(\frac{V_b + K}{V_d + K} \right)^r, \quad V_d \geq V_b \quad (\text{S10})$$

This distribution is *monotonically decreasing* on $[V_b, \infty)$: its logarithmic derivative is $-(r+1)/(V_d + K) < 0$ everywhere. The mode is at the boundary $V_d = V_b$, not at an interior point.

The **mean** division volume is computed exactly:

$$\langle V_d \rangle = (V_b + K) \cdot \frac{r}{r-1} - K = \frac{V_b + K}{r-1} \quad (\text{for } r > 1) \quad (\text{S11})$$

giving $\langle \Delta V \rangle = (V_b + K)/(r-1)$.

The **median** satisfies $S(V_d^{\text{med}}) = 1/2$:

$$V_d^{\text{med}} = (V_b + K) \cdot 2^{1/r} - K \quad (\text{S12})$$

For $K \gg V_b$, both converge to constants independent of V_b :

$$\langle \Delta V \rangle \approx \frac{K}{r-1} \left[1 + \frac{V_b}{K} + O\left(\frac{V_b^2}{K^2}\right) \right] \quad (\text{S13})$$

which is the **pure adder**: added volume independent of birth size to leading order.

Error estimate: The relative deviation from perfect adder is $|\Delta V(V_b) - \Delta V(0)|/\Delta V(0) = V_b/K$. For *E. coli*, $\langle V_b \rangle \approx 1.5 \mu\text{m}^3$ and $K \approx 15 \mu\text{m}^3$ (from fits), giving $\sim 10\%$ deviation—consistent with the experimentally observed weak residual slope in the $(V_b, \Delta V)$ plot.

S4 Transient overshoot derivation (Prediction 1)

After a nutrient upshift at $t = 0$, μ jumps from μ_1 to $\mu_2 > \mu_1$, while K adjusts on a slower timescale τ_K :

$$K(t) = K_2 - (K_2 - K_1)e^{-t/\tau_K} \quad (\text{S14})$$

Since the hazard scales with μ , the growth and division rates both increase immediately. However, if $K_2 > K_1$ (faster growth requires larger cells, as observed empirically), K lags behind.

Proposition 3 (Overshoot condition). *A transient overshoot in mean cell size occurs when:*

$$\tau_K > \frac{1}{\mu_2} \cdot \frac{n}{r} \cdot \frac{K_2}{K_2 - K_1} \quad (\text{S15})$$

The overshoot magnitude is:

$$\frac{\langle V \rangle_{\max} - \langle V \rangle_{ss}}{\langle V \rangle_{ss}} \approx \frac{K_2 - K_1}{K_2} \cdot (1 - e^{-\mu_2 \tau_K}) \quad (\text{S16})$$

The overshoot arises because the effective division size temporarily decreases (old K) while cells grow faster (new μ), creating a mismatch that resolves non-monotonically as $K(t) \rightarrow K_2$.

S5 Phase-specific control combination rule (Prediction 3)

For a cell cycle divided into two phases (e.g., G1 and S/G2/M), each governed by its own hazard function $h_i(V) = \mu r_i V^{n_i} / (V^{n_i} + K_i^{n_i})$, the whole-cycle effective behaviour depends on the sequential application of the two hazard functions.

The effective whole-cycle Amir slope α_{eff} (where $V_d = (1 - \alpha)V^* + \alpha V_b + \Delta V$) obeys a multiplicative combination rule:

$$\alpha_{\text{eff}} = \alpha_1 \cdot \alpha_2 \quad (\text{S17})$$

where $\alpha_i = 1 - n_i / (r_i + n_i)$ is the phase-specific slope.

Derivation. In phase 1, the conditional expectation of the transition volume V_t (volume at which the cell exits phase 1) given birth volume V_b has the approximate form $\langle V_t \rangle = (1 - \alpha_1)V_1^* + \alpha_1 V_b$ where V_1^* is the

phase-1 target size. Phase 2 then acts on V_t as birth size, giving $\langle V_d \rangle = (1 - \alpha_2)V_2^* + \alpha_2\langle V_t \rangle$. Substituting:

$$\langle V_d \rangle = (1 - \alpha_2)V_2^* + \alpha_2(1 - \alpha_1)V_1^* + \alpha_1\alpha_2V_b \quad (\text{S18})$$

so the whole-cycle slope is $\alpha_{\text{eff}} = \alpha_1\alpha_2$.

Application to budding yeast. G1 operates as a “weak sizer” with $n_1 = 3-5$, giving $\alpha_1 \approx 0.3-0.5$. Post-Start operates as a timer ($n_2 \rightarrow 0$, $\alpha_2 = 1$). Therefore $\alpha_{\text{eff}} \approx 0.3-0.5$, consistent with the observed $\alpha \approx 0.4$ for whole-cycle budding yeast[?].

For G1 pure sizer ($n_1 \gg 1 \Rightarrow \alpha_1 \rightarrow 0$) plus any phase 2: $\alpha_{\text{eff}} \rightarrow 0$ (whole-cycle sizer). For both phases timer ($n_1, n_2 \rightarrow 0$): $\alpha_{\text{eff}} = 1 \cdot 1 = 1$ (whole-cycle timer, slope 2 in exponential growth).

S6 Population-level dynamics and moment equations

S6.1 Structured population equation (Fokker–Planck)

Let $\phi(V, t)$ be the volume density of cells at time t . The population evolves according to:

$$\frac{\partial \phi}{\partial t} + \frac{\partial}{\partial V}(\mu V \phi) = -h(V)\phi + 4\mu r \frac{(2V)^n}{(2V)^n + K^n} \phi(2V, t) \quad (\text{S19})$$

where the first term on the right is loss from division at rate $h(V)$, and the second is gain from mothers of size $2V$ dividing symmetrically. At steady state, this reduces to the renewal equation in Section S2, confirming internal consistency.

This population-level PDE is the deterministic “master equation” of the system. The hazard function $h(V)$ encodes all control information: the framework is a stochastic process at the single-cell level whose population statistics are governed by a structured population equation.

S6.2 Moment equations and the two-parameter reduction

Let $\rho_g(V_b)$ be the birth-size density in generation g . The map $V_b^{(g+1)} = V_d^{(g)}/2$ gives:

$$\rho_{g+1}(V_b') = 2 \int_0^\infty p(2V_b' | V_b) \rho_g(V_b) dV_b \quad (\text{S20})$$

At steady state, the condition $\langle V_d \rangle = 2\langle V_b \rangle$ (cells must double on average for population stability) con-

strains r as a function of n and $K/\langle V_b \rangle$. For $n = 1$: $r = 1 + K/\langle V_b \rangle \cdot (1/\ln 2)$.

This explains why the model has effectively **two free control parameters** (n, K) despite having three: r is determined by the steady-state constraint. The “two-parameter family” refers to the (n, K) plane that defines the phase diagram of control strategies, where n sets the *type* of control (cooperativity) and K sets the *target size* (inhibitor dosage). The third parameter r controls *precision* (fidelity of control) and is fixed by the doubling requirement.

S7 Supplementary Figures

S8 Supplementary Tables